

Education

Department of Biological Sciences, University of Pittsburgh

PH.D. CANDIDATE IN MOLECULAR, CELLULAR, AND DEVELOPMENTAL BIOLOGY

Pittsburgh, PA, USA

Aug 2021 – Present

- Advisor: **Dr. Jacob Durrant**
- Expected graduation is December 2027

University of Science and Technology, Zewail City

B.S. IN COMPUTATIONAL BIOLOGY AND GENOMICS (BIOMEDICAL SCIENCE)

Giza, Egypt

Aug 2017 – Jul 2021

- Advisor: **Dr. Eman Badr**
- Graduated *cum laude*; awarded a **90% merit-based scholarship** throughout all four years of study.

Research Experience

Graduate Student Researcher

Pittsburgh, PA, USA

DEPARTMENT OF BIOLOGICAL SCIENCES, UNIVERSITY OF PITTSBURGH

Sep 2021 – Present

- **LIGNOVA: protein–ligand complex generation pipeline:** Built a fully automated, open-source pipeline that docks bioactive compounds against experimentally-determined protein structures. Processing just ~20% of the data has already yielded over **1.2 million protein–ligand complexes** and is on track to produce ~6 million—an order-of-magnitude increase over existing structural data for drug design and ML training.
- **Water modeling in protein–ligand docking:** Discovered a **hydration-dependent transition** between two scoring regimes by benchmarking 255 crystal structures across three hydration conditions: when water is absent, consensus-based rescoring recovers ~30% of failed poses given sufficient ligand data, but when explicit water is retained, the consensus signal collapses and the physics-based docking score alone is sufficient. Provides actionable guidance on when expensive water modeling can be safely omitted. **Manuscript in preparation.**
- **Structure-based discovery of arginine sensors in yeast (Collaborative with O'Donnell Lab):** Identified **4 candidate arginine-sensing proteins** in *S. cerevisiae* that are invisible to traditional sequence homology approaches by combining Foldseek structural search across 5 human CASTOR1 structures with **MolModa**. Experiments confirmed that identified candidates show altered arginine homeostasis phenotypes.
- Built and maintained Python pipelines with **HPC automation** (SLURM) for large-scale structural and molecular data processing across all projects.

Undergraduate Student Researcher

Giza, Egypt

UNIVERSITY OF SCIENCE AND TECHNOLOGY, ZEWAİL CITY

Jun 2020 – Apr 2021

- Trained Random Forest regression models (scikit-learn) to predict **drug sensitivity from transcriptomic profiles** in acute myeloid leukemia patients, identifying genetic markers correlated with differential drug response.

Publications

Ahmed, M.; Maldonado, A. M.; Durrant, J. D. From hydration to simplicity: Leveraging nonstructural data for protein–ligand docking without explicit water modeling. **In preparation.**

Holland, M.*; **Ahmed, M.***; Young, J. M.; Drurey, J. R.; McFadyen, S.; Ostrowski, E. A.; Levin, T. C. Hypermutable hotspot enables the rapid evolution of self/non-self recognition genes in *Dictyostelium*. *PNAS* **2025**, 122(51). DOI: 10.1073/pnas.2520843122. (*co-first authors)

Kochnev, Y.; **Ahmed, M.;** Maldonado, A. M.; Durrant, J. D. MolModa: Accessible and secure molecular docking in a web browser. *Nucleic Acids Research* **2024**, 52(W1), W498–W506. DOI: 10.1093/nar/gkae406.

Ahmed, M.; Maldonado, A. M.; Durrant, J. D. From byte to bench to bedside: Molecular dynamics simulations and drug discovery. *BMC Biology* **2023**, 21(1), 299. DOI: 10.1186/s12915-023-01791-z.

Conferences

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|------|--|----------------|
| 2024 | Poster Presenter — Beyond the Blueprint: A Novel Database for Innovative Drug Design and Discovery, Gordon Research Conference & Seminar on Computational Chemistry | Portland, ME |
| 2023 | Poster Presenter — Augmenting Protein–Ligand Complex Databases with PubChem for Enhanced Drug Discovery, Gordon Research Conference & Seminar on Computer-Aided Drug Design | West Dover, VA |

Skills

Programming	Python, Bash, R
Computational Tools	Git, GitHub, Linux/SLURM, HPC cluster management
Machine Learning	Supervised Learning, Feature Engineering, Model Evaluation, Graph Neural Networks
Data Analysis	Statistical Modeling, Data Visualization (Matplotlib, Seaborn)
Packages	PyTorch, scikit-learn, NumPy, pandas, Polars, SciPy, RDKit, Biopython, MDAnalysis, PyArrow
Molecular Modeling	Docking (Glide, GNINA, AutoDock Vina), Structural Search (Foldseek), Preparation (PDB2PQR, Gypsum-DL)
Languages	English (Fluent), Arabic (Native)

Teaching & Mentorship Experience

Undergraduate Research Mentor

TECBIO REU, DEPARTMENT OF COMPUTATIONAL & SYSTEMS BIOLOGY, UNIVERSITY OF PITTSBURGH

Pittsburgh, PA, USA

May 2024 – Jul 2024

- Co-mentored an undergraduate student on a 10-week project applying AlphaFold2 to explore HIV protease conformations for drug design.
- Supervised development of a research poster, presented at the Research Symposium at Duquesne University.

Graduate Teaching Assistant & Fellow (Biochemistry)

DEPARTMENT OF BIOLOGICAL SCIENCES, UNIVERSITY OF PITTSBURGH

Pittsburgh, PA, USA

Spring 2024, Fall 2025

- Conducted recitation sections and held weekly office hours for ~200 students; mentored 15 undergraduate teaching assistants.
- Nominated for the **Elizabeth Baranger Excellence in Teaching Award** based on outstanding student evaluations.

Undergraduate Teaching Assistant (Python)

UNIVERSITY OF SCIENCE AND TECHNOLOGY, ZEWAIL CITY

Giza, Egypt

Sep 2020 – Jan 2021

- Guided students in writing and debugging Python code; held weekly office hours.

Leadership & Service

Pitt Biological Sciences Graduate Student Organization

ARTS & SCIENCES GRADUATE STUDENT ORGANIZATION REPRESENTATIVE

Pittsburgh, PA, USA

Aug 2024 – Present

- Represent Biological Sciences at school-wide GSO meetings, addressing department concerns.
- Coordinate with the GSO executive board on event planning, policy discussions, and decision-making processes.
- **Served on the Summer Research Grant panel**, evaluating graduate student research proposals across different A&S departments.

Pitt Biological Sciences Graduate Student Organization

GRADUATE DIVERSITY, INCLUSION, AND BELONGING (DIB) COMMITTEE MEMBER

Pittsburgh, PA, USA

Sep 2023 – Dec 2024

- DIB works on fostering an inclusive environment for underrepresented students in the department.
- Organize intercultural celebrations and distribute informational resources highlighting the diverse cultural traditions within the department.

Zewail City Opencourseware

VICE PRESIDENT

Egypt

Jan 2018 – Mar 2020

- Led a team making university course materials (lecture notes, videos, assignments) freely available online for Egyptian students.
- Selected courses to document each semester, coordinated sub-teams, and contributed LaTeX-typeset lecture notes and solutions.

Bio Team Egypt

HEAD OF STRATEGY COMMITTEE

Egypt

Jan 2017 – Sep 2017

- Managed 3 subcommittees (Events, Awareness Campaigns, Charities) and organized 3 biotechnology seminars.
- Coordinated 4 public awareness campaigns introducing biotechnology to high school students across Egypt.